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B.Sc. IT (Hons.)
Artificial Intelligence

Talk2MeLatin

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INTRODUCTION

In recent years, the use of AI in education, particularly in language learning, has been growing, with conversational interfaces enabling learners to engage in interactive dialogue [1] . Despite these advances, classical languages such as Latin do not benefit from the same amount of AI-supported educational tools that several modern languages do.

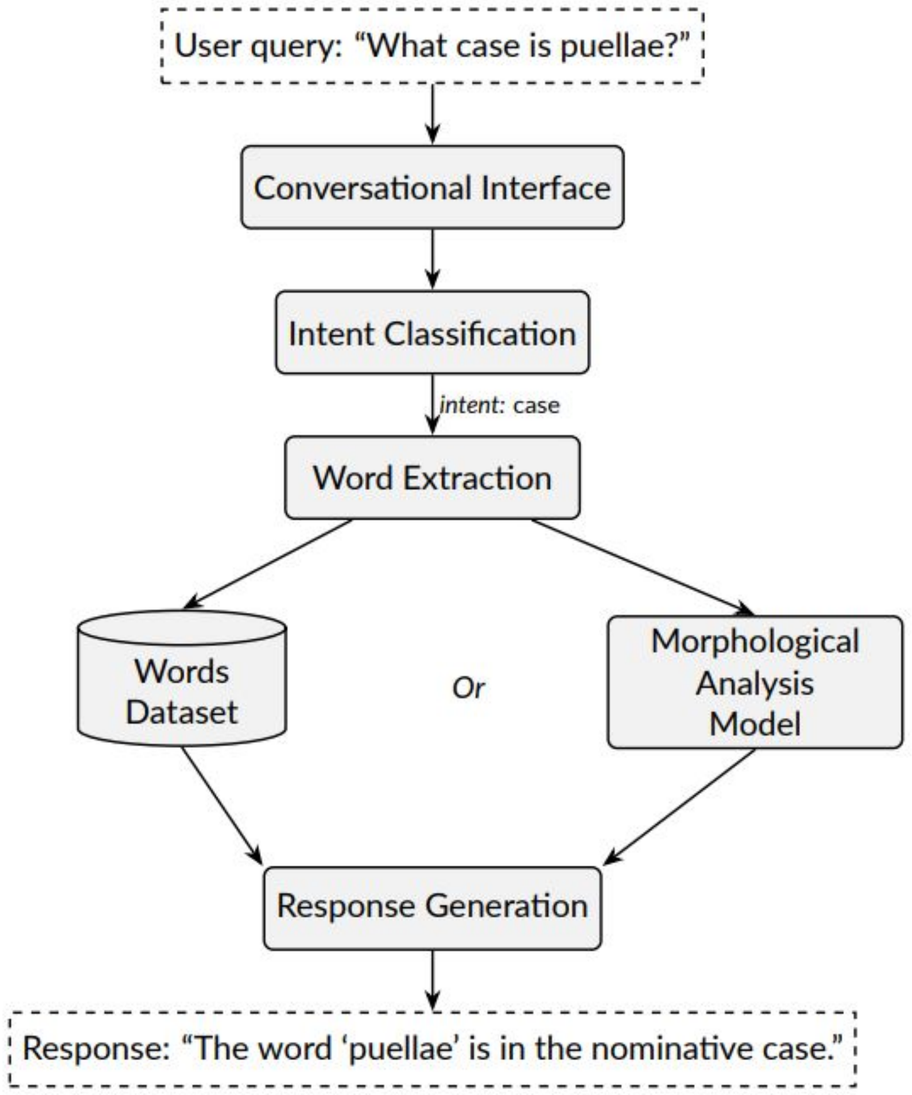
This project addresses this by developing a conversational system that:
- Takes as input a natural-language question about the morphological properties of a Latin word.
- Interprets user intent.
- Outputs a natural-language response that identifies which grammatical aspect the question targets and provides the morphological information for the Latin word in question.

AIM

The main aim was to develop a conversational tool for Latin language learners to provide answers to questions related to the morphological features of a Latin word.

This was achieved through four primary objectives:
- Construct two datasets: (i) a dataset of Latin words and their grammatical features; (ii) a synthetic dataset of grammar questions for training the intent classification model.
- Fine-tune two BERT-based models; one for intent classification, and another for morphological analysis.
- Develop a response generation component that combines the intent and morphological information into an answer, accessible through a conversational interface.
- Evaluate the system through quantitative metrics and qualitative user testing with Latin learners.

ARCHITECTURE DESIGN



METHODOLOGY

Constructed the datasets required by the system. → **Fine-tuned a BERT [2] model for intent classification to identify user queries.** → **Fine-tuned Latin BERT [3] for morphological analysis of a Latin wordform.** → **Developed a conversational UI that returns a natural-language response.**

RESULTS

Results demonstrated that the intent classifier achieved an overall accuracy and F1 score of 0.97, while the morphological analyser achieved per-feature accuracies ranging from 0.71 to 0.90, with a full-tag accuracy of 0.56. Furthermore, for many individual features of the morphological analyser, a large gap between macro and weighted F1 scores were noted, indicating that the model performs poorly on minority classes, and achieves higher weighted F1 and accuracy scores by just predicting the dominant class.

A user evaluation revealed that, despite high test-set performance of the intent classifier, the system struggled to understand many real learner questions and was perceived as having limited pedagogical usefulness. The contrast between the technical and user evaluations highlighted a gap between the templated training distribution and the variety of possible questions actual learners ask.

CONCLUSIONS AND FUTURE WORK

This project presents the development of a prototype conversational chatbot that answers Latin grammar questions by combining an intent classifier and a morphological analyser, making a step towards the development of a conversational tool that can help Latin learners.

Future directions include:
- More diverse training data for the intent classifier.
- Extend the types of questions the system can handle to include additional capabilities.
- Improve the performance of the morphological analyser on minority classes.
- Revise the Latin words dataset to prioritise Classical Latin forms.

REFERENCES

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